

Development of a Wireless Active Personal Dosimeter for Interventional Radiology

M. A. Moniz¹, P. M. M. Correia¹, A. L. M. Silva¹, J. F. C. A. Veloso¹, C. D. R. Azevedo¹

¹i3N, Departamento de Física, Universidade de Aveiro, 3810-193 Aveiro, Portugal

Abstract:

INTRODUCTION

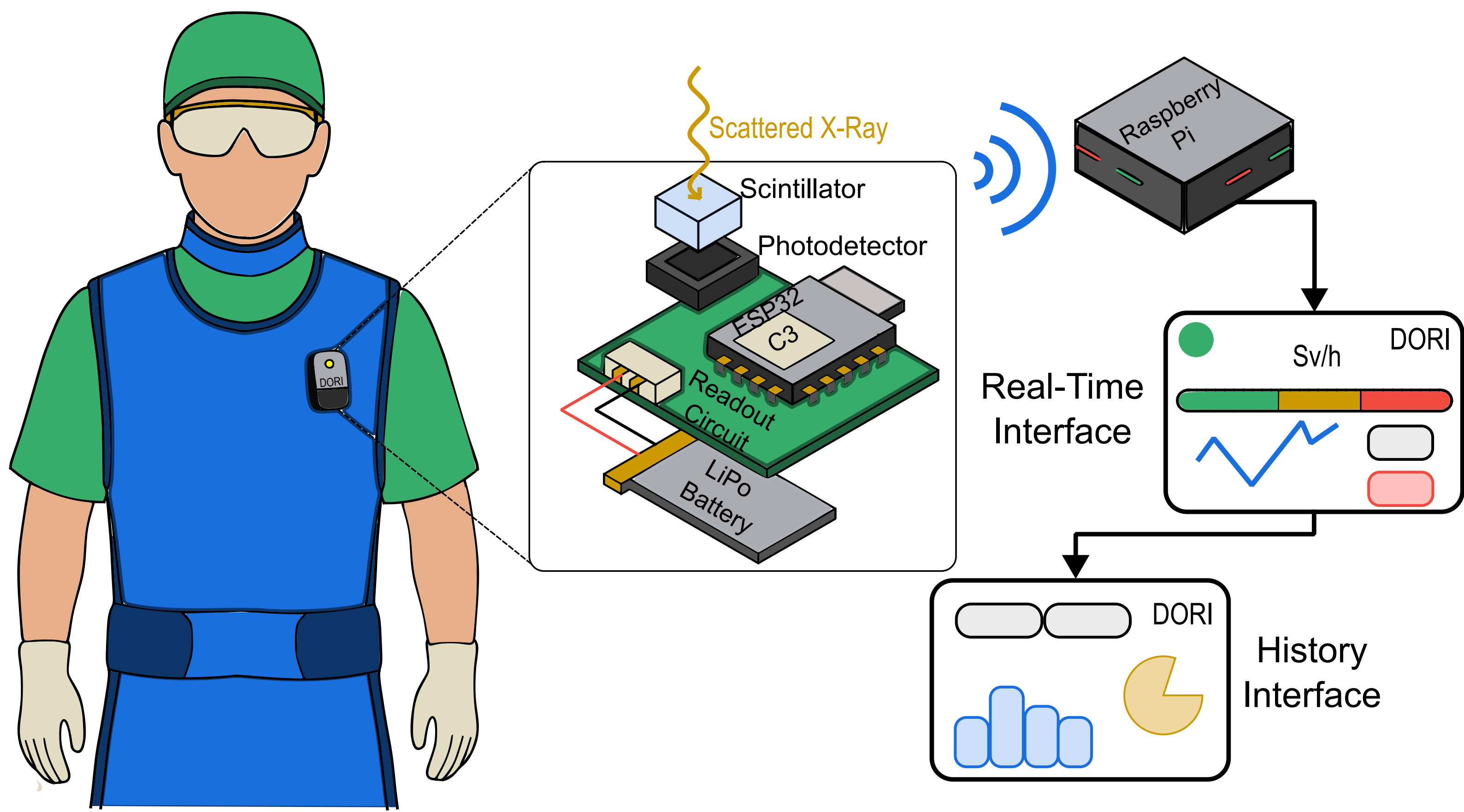


Figure 1. Conceptual design of DORI.

METHODS

Hardware

Energy Calibration

Dose Calibration

RESULTS

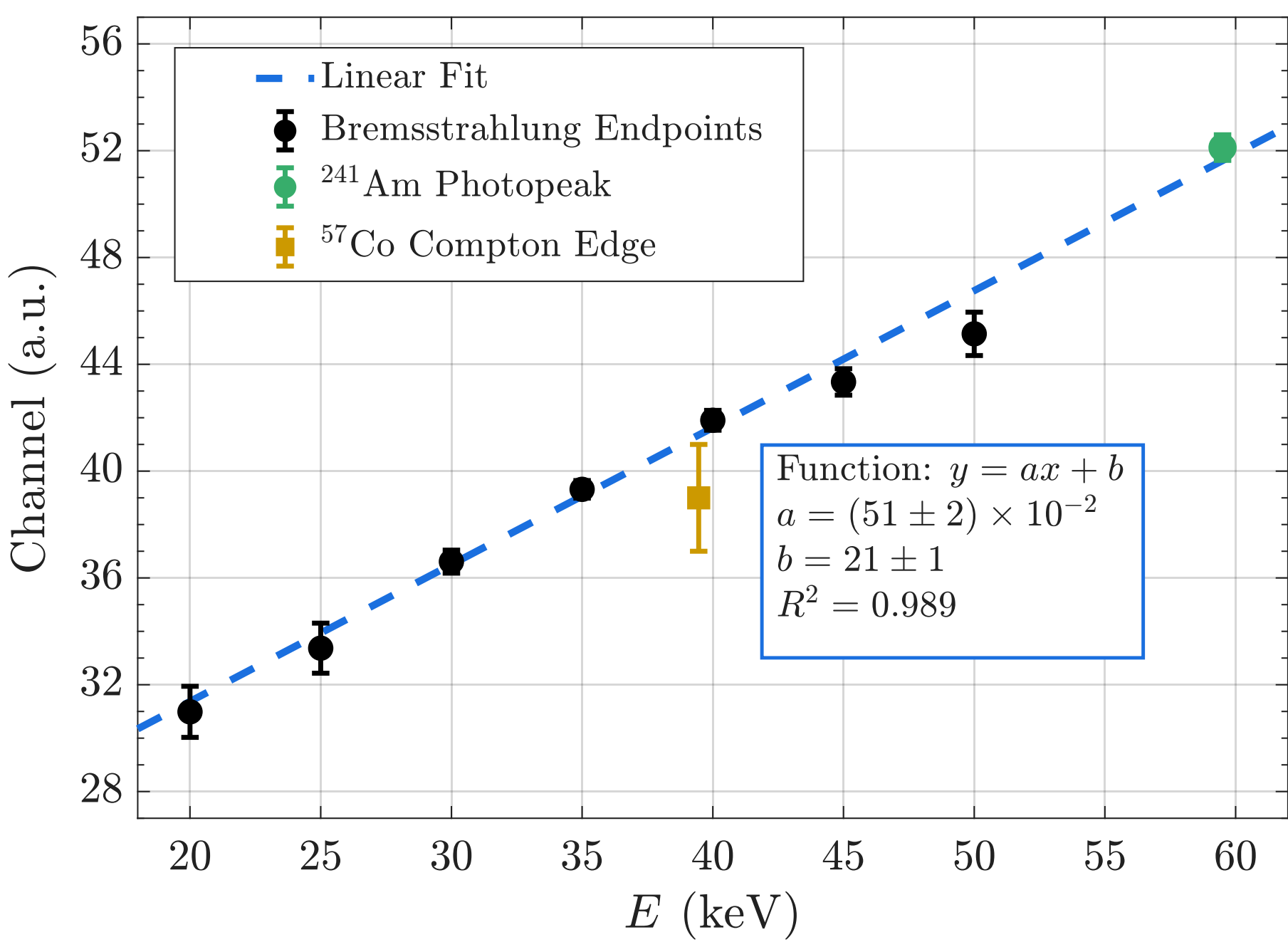


Figure 2. Energy calibration.

DISCUSSION

CONCLUSIONS

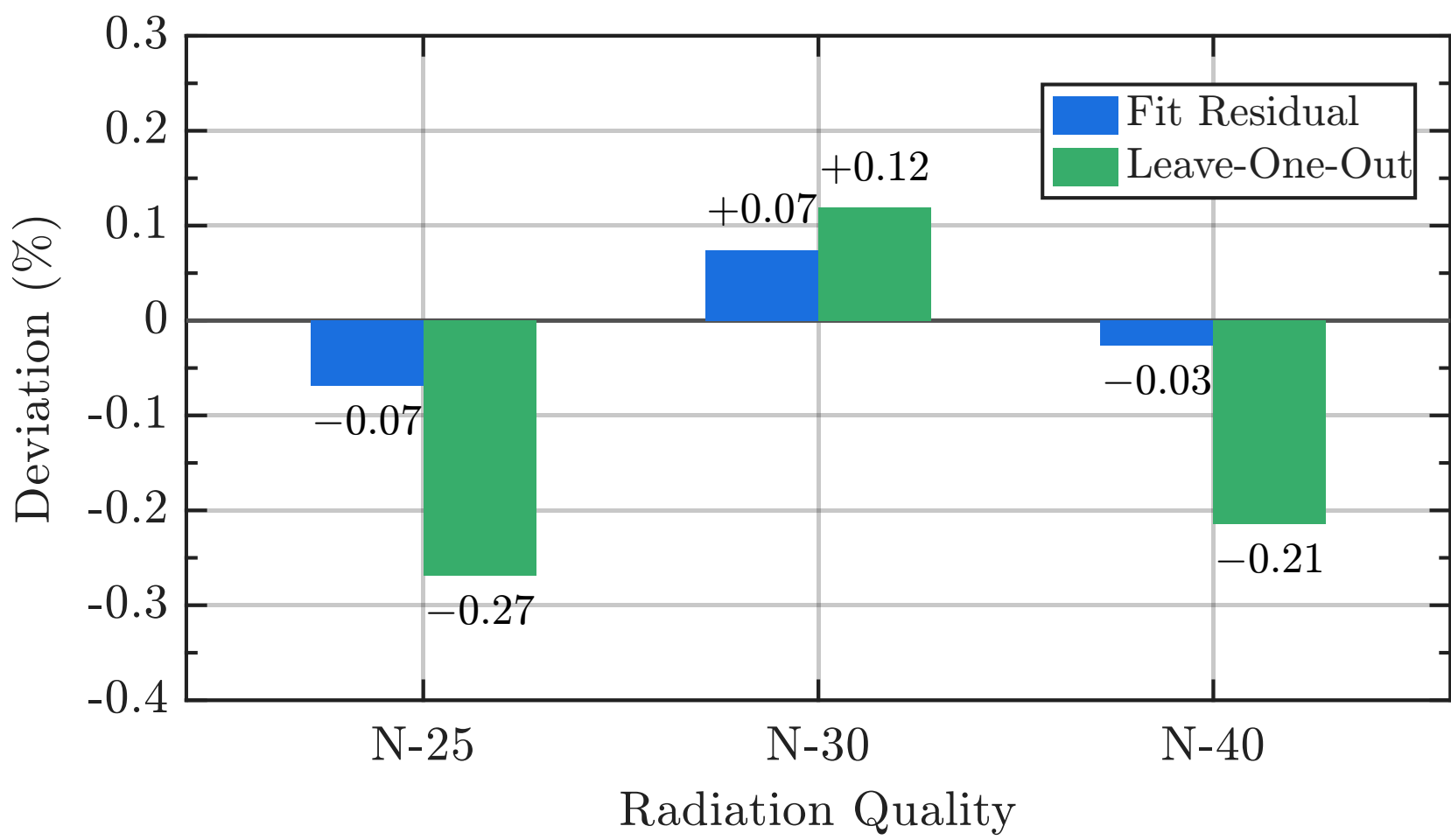


Figure 3. Dose calibration.

Table 1. Characteristics of the DORI prototype.

Parameter	Value
Dead time	$17.8 \pm 0.4 \mu\text{s}$
Energy range	20 to 60 keV
Operational unit	$H_p(10)$
Dose range	N-25 to N-40
Dimensions	$6.3 \times 7.2 \times 2.5 \text{ cm}$
Mass	141 g
Autonomy	8.6 h

REFERENCES